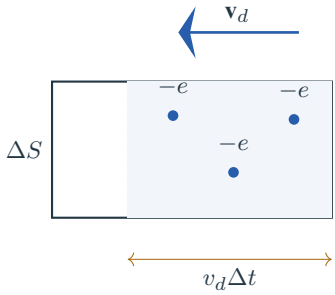
穿过截面的电荷来自一段扫掠体积



$$\Delta N = n v_d \Delta t \, \Delta S$$

$$j = e\Delta N/(\Delta t \Delta S) = n e v_d$$

$$\mathbf{J} = -n e \mathbf{v}_d$$

v_d 为漂移速率；矢量方向最后由负电荷决定